

Q1:

1. Cells mean:

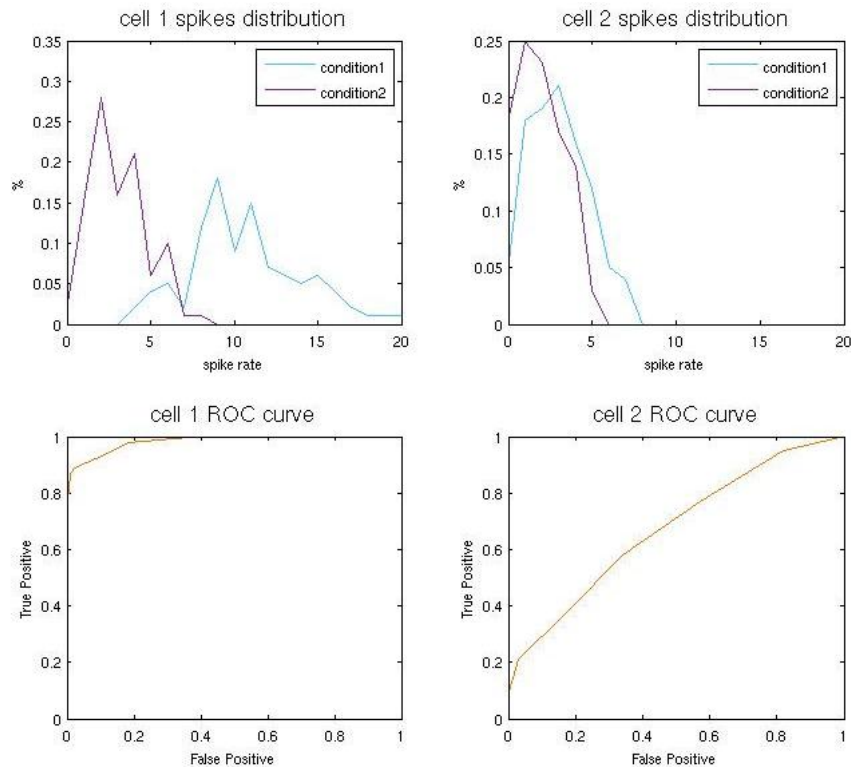
cell1, condition1: 10.6200

cell1, condition2: 3.0800

cell2, condition1: 3.0100

cell2, condition2: 1.9300

2.



Top subplot represents cells 1&2 in two conditions: Condition1: cat presented ; Condition2: cat not presented.
Bottom subplots represent cells 1&2 ROC curves, cell 1 has an area of 0.9839, cell 2 has an area of 0.6716

3. By the ROC curves we can clearly observe that cell 1 has a much better predictive ability as the area under it is close to 1, which represents a perfect separation. Cell 2's ROC curve is close to 0.5, a random discrimination. The cells' spikes distribution graphs visualize it, as there's higher intersection between the two conditions in cell 2 compared to that of cell 1's.

4. Cell 1's ROC curve has an area of 0.9839

Cell 2's ROC curve has an area of 0.6716

5.

$$\lambda_{10} = 1, \lambda_{01} = 50, \lambda_{11} = \lambda_{00} = 0$$

$$P(y = 1) = 0.04 \rightarrow P(y = 0) = 1 - P(y = 1) = 1 - 0.04 = 0.96$$

$$\text{from matlab: } P(x = 6|y = 1) = 0.05, \quad P(x = 6|y = 0) = 0.1$$

$$\begin{aligned} R(\alpha = 1|x = 6) &= \text{error}(\alpha = 1|y = 0) \times P(y = 0|x = 6) + \text{error}(\alpha = 1|y = 1) \times P(y = 1|x = 6) \\ &= \lambda_{10} \frac{P(x = 6|y = 0) \times P(y = 0)}{P(x = 6|y = 0) \times P(y = 0) + P(x = 6|y = 1) \times P(y = 1)} \\ &\quad + \lambda_{11} \frac{P(x = 6|y = 1) \times P(y = 1)}{P(x = 6|y = 0) \times P(y = 0) + P(x = 6|y = 1) \times P(y = 1)} \\ &= 1 \times \frac{0.1 \times 0.96}{0.1 \times 0.96 + 0.05 \times 0.04} + 0 = 0.9795 \end{aligned}$$

$$\begin{aligned} R(\alpha = 0|x = 6) &= \text{error}(\alpha = 0|y = 0) \times P(y = 0|x = 6) + \text{error}(\alpha = 0|y = 1) \times P(y = 1|x = 6) \\ &= \lambda_{00} \frac{P(x = 6|y = 0) \times P(y = 0)}{P(x = 6|y = 0) \times P(y = 0) + P(x = 6|y = 1) \times P(y = 1)} \\ &\quad + \lambda_{01} \frac{P(x = 6|y = 1) \times P(y = 1)}{P(x = 6|y = 0) \times P(y = 0) + P(x = 6|y = 1) \times P(y = 1)} \\ &= 0 + 50 \times \frac{0.05 \times 0.04}{0.05 \times 0.04 + 0.1 \times 0.96} = 1.0204 \end{aligned}$$

Given that the cell fired 6 times, it is preferred to run away.

$$6. \quad \lambda_{11} = V, \lambda_{10} = V + 1$$

נדרוש:

$$LR = \frac{R(\alpha = 1)}{R(\alpha = 0)} > 1$$

$$\begin{aligned} R(\alpha = 1) &= \text{error}(\alpha = 1|y = 0) \times P(y = 0|x = 6) + \text{error}(\alpha = 1|y = 1) \times P(y = 1|x = 6) \\ &= \lambda_{10} \frac{P(x = 6|y = 0) \times P(y = 0)}{P(x = 6|y = 0) \times P(y = 0) + P(x = 6|y = 1) \times P(y = 1)} \\ &\quad + \lambda_{11} \frac{P(x = 6|y = 1) \times P(y = 1)}{P(x = 6|y = 0) \times P(y = 0) + P(x = 6|y = 1) \times P(y = 1)} \\ &= (V + 1) \times 0.9795 + V \times 0.0204 = 0.999V + 0.9795 \end{aligned}$$

$$\begin{aligned} R(\alpha = 0) &= \text{error}(\alpha = 0|y = 0) \times P(y = 0|x = 6) + \text{error}(\alpha = 0|y = 1) \times P(y = 1|x = 6) \\ &= \lambda_{00} \frac{P(x = 6|y = 0) \times P(y = 0)}{P(x = 6|y = 0) \times P(y = 0) + P(x = 6|y = 1) \times P(y = 1)} \\ &\quad + \lambda_{01} \frac{P(x = 6|y = 1) \times P(y = 1)}{P(x = 6|y = 0) \times P(y = 0) + P(x = 6|y = 1) \times P(y = 1)} = 1.0204 \end{aligned}$$

$$\frac{0.999V + 0.9795}{1.0204} > 1 \rightarrow 0.999V + 0.9795 > 1.0204 \rightarrow V > 0.0413$$

Q2:

$$P(T = 1) = 0.01$$

$$P(T = 0) = 1 - P(T = 1) = 1 - 0.01 = 0.99$$

$$P(X = 0|T = 1) = 0.2$$

$$P(X = 1|T = 0) = 0.096$$

1.

	X=0	X=1	
T=0	0.89496	0.09504	0.99
T=1	0.002	0.008	0.01
	0.89696	0.10304	

$$P(X = 1, T = 0) = P(X = 1|T = 0) \times P(T = 0) = 0.096 \times 0.99 = 0.09504$$

$$P(X = 0, T = 0) = P(T = 0) - P(X = 1, T = 0) = 0.99 - 0.09504 = 0.89496$$

$$P(X = 0, T = 1) = P(X = 0|T = 1) \times P(T = 1) = 0.2 \times 0.01 = 0.002$$

$$P(X = 1, T = 1) = P(T = 1) - P(X = 0, T = 1) = 0.01 - 0.002 = 0.008$$

$$2. P(T = 1|X = 1) = \frac{P(X=1,T=1)}{P(X=1)} = \frac{0.008}{0.10304} = 0.10304$$

$$3. P(T = 0|X = 0) = \frac{P(X=0,T=0)}{P(X=0)} = \frac{0.89496}{0.89696} = 0.99777025$$

$$4. P(T = 1|X = 1) = 0.5$$

$$\frac{P(T=1,X=1)}{P(X=1)} = 0.5$$

$$P(X = 1) = \frac{P(T=1,X=1)}{0.5} = \frac{0.008}{0.5} = 0.016$$

$$P(X = 1|T = 0) = \frac{P(T=0|X=1) \times P(X=1)}{P(T=0)} = \frac{0.5 \times 0.016}{0.99} = \frac{4}{495}$$

	X=0	X=1	
T=0	0.982	0.008	0.99
T=1	0.002	0.008	0.01
	0.984	0.016	